



Module 5: Data Visualization

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Contents

Data Visualization:

- Basic principles,
- Ideas and tools for data visualization

Data Visualization

- Data visualization is the graphical representation of information and data, using visual elements like charts, graphs, maps, and dashboards to make complex datasets understandable, actionable, and accessible. It transforms raw data into insights, highlighting trends, correlations, and outliers that might otherwise remain hidden.

Data Visualization

What is Data Visualization?

Turning data into visual formats like charts, graphs, and maps

Makes complex data easy to understand at a glance

Helps identify trends, patterns, and insights



Key Concepts of Data Visualization

- **Types of Visualizations:** Common forms include bar charts, line graphs, pie charts, scatter plots, maps, and interactive dashboards.
- **Purpose:** The primary goal is to communicate information clearly and efficiently, allowing users to analyze data and make data-driven decisions.

Key Concepts of Data Visualization

Types of Analysis:

- **Exploratory:** Used to dig into datasets, uncover hidden patterns, and identify areas needing deeper investigation.
- **Explanatory:** Used to tell a specific story or convey a clear message to an audience.
- **Benefits:** It helps in identifying trends (e.g., climate change patterns), spotting anomalies, and speeding up decision-making.

Advantages of Data Visualization

Advantages of Data Visualization

- Easily sharing information.
- Interactively explore opportunities.
- Visualize patterns and relationships.

Advantages of Data Visualization



Disadvantages of Data Visualization

Disadvantages of Data Visualization

- Biased or inaccurate information.
- Correlation doesn't always mean causation.
- Core messages can get lost in translation.

Disadvantages of Data Visualization



Advantages And Disadvantages of Data Visualization

Advantages



Executive Clarity



**Faster Alignment
Across Teams**



Early Exception Detection



**Better Decision
Communication**



**Scalable Performance
Visibility**

VS

Disadvantages



False Confidence Risk



**Metric Conflicts
Across Teams**



**Oversimplification Of
Drivers**



Adoption Without Action



**Security And
Exposure Risk**

Why Data Visualization is Important?

- The importance of data visualization is simple: it helps people see, interact with, and better understand data.
- Whether simple or complex, the right visualization can bring everyone on the same page, regardless of their level of expertise.

Why Data Visualization is Important?

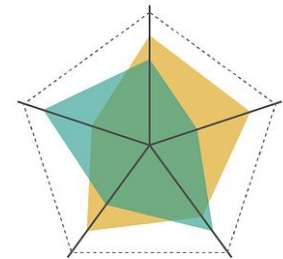
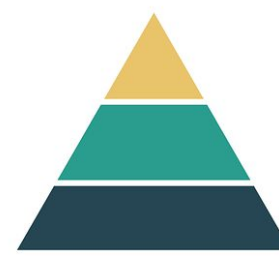
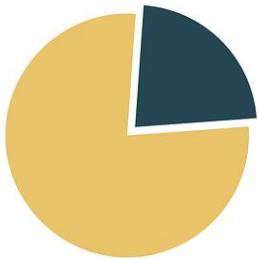
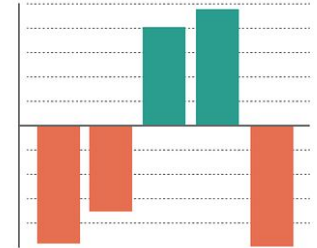
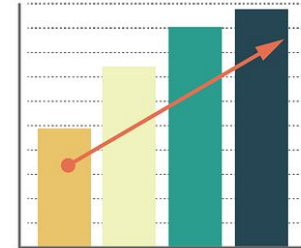
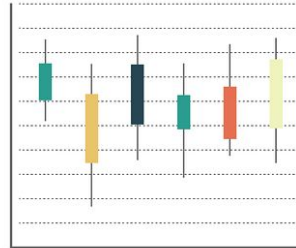
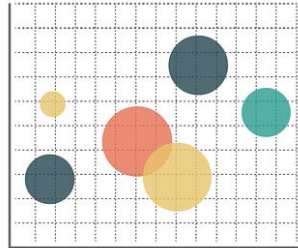
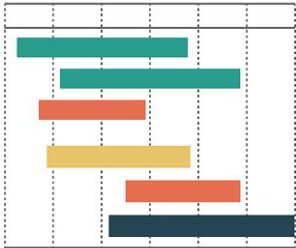
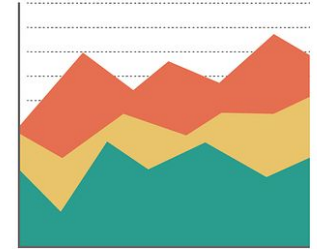
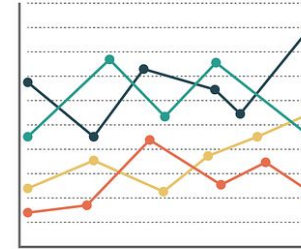
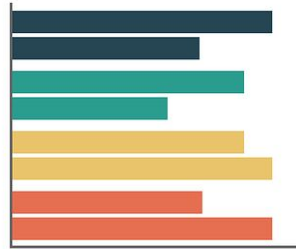
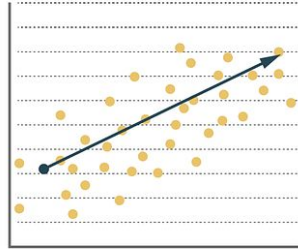
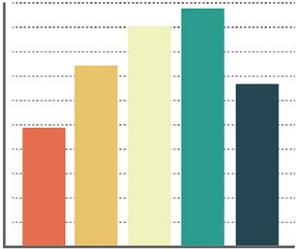
Importance of Data visualization



Types of Data Visualization

- Chart: Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map.
- Table: A set of figures displayed in rows and columns.
- Graph: A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other, usually along two axes at a right angle.

Types of Data Visualization

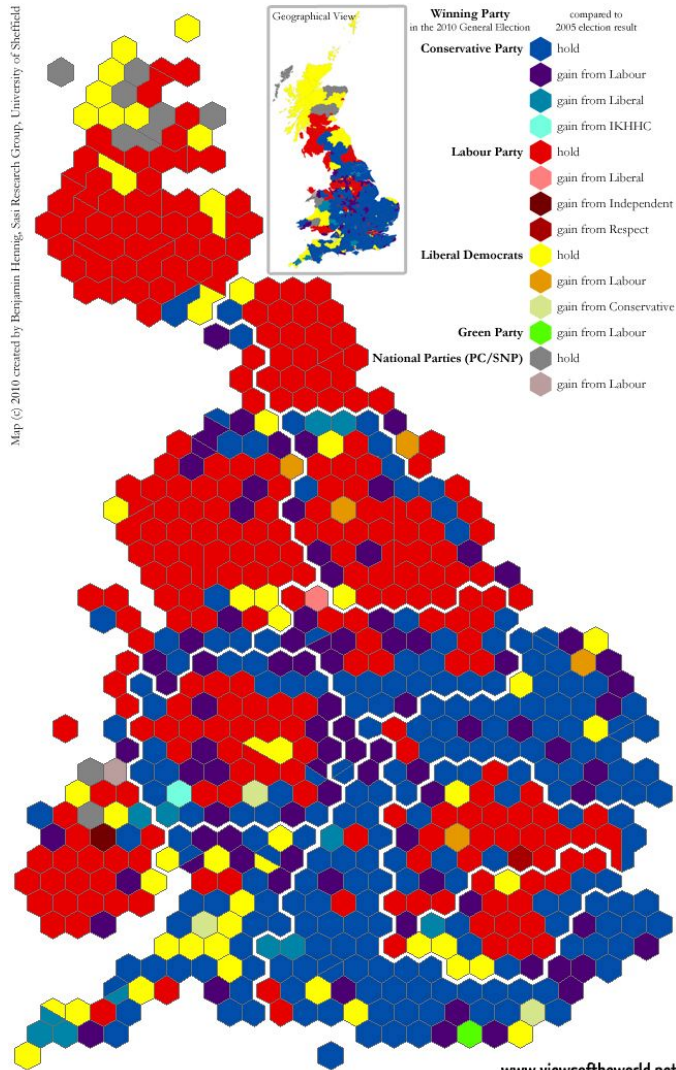


Types of Data Visualization

- **Geospatial:** A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations.
- **Infographic:** A combination of visuals and words that represent data. Usually uses charts or diagrams.
- **Dashboards:** A collection of visualizations and data displayed in one place to help with analyzing and presenting data.

Geospatial

Map (c) 2010 created by Benjamin Herwig, SAS Research Group, University of Sheffield



Other Examples

- Area Map: A form of geospatial visualization, area maps are used to show specific values set over a map of a country, state, county, or any other geographic location. Two common types of area maps are choropleths and isopleths.
- Bar Chart: Bar charts represent numerical values compared to each other. The length of the bar represents the value of each variable.

Other Examples

- Box-and-whisker Plots: These show a selection of ranges (the box) across a set measure (the bar).
- Bullet Graph: A bar marked against a background to show progress or performance against a goal, denoted by a line on the graph.

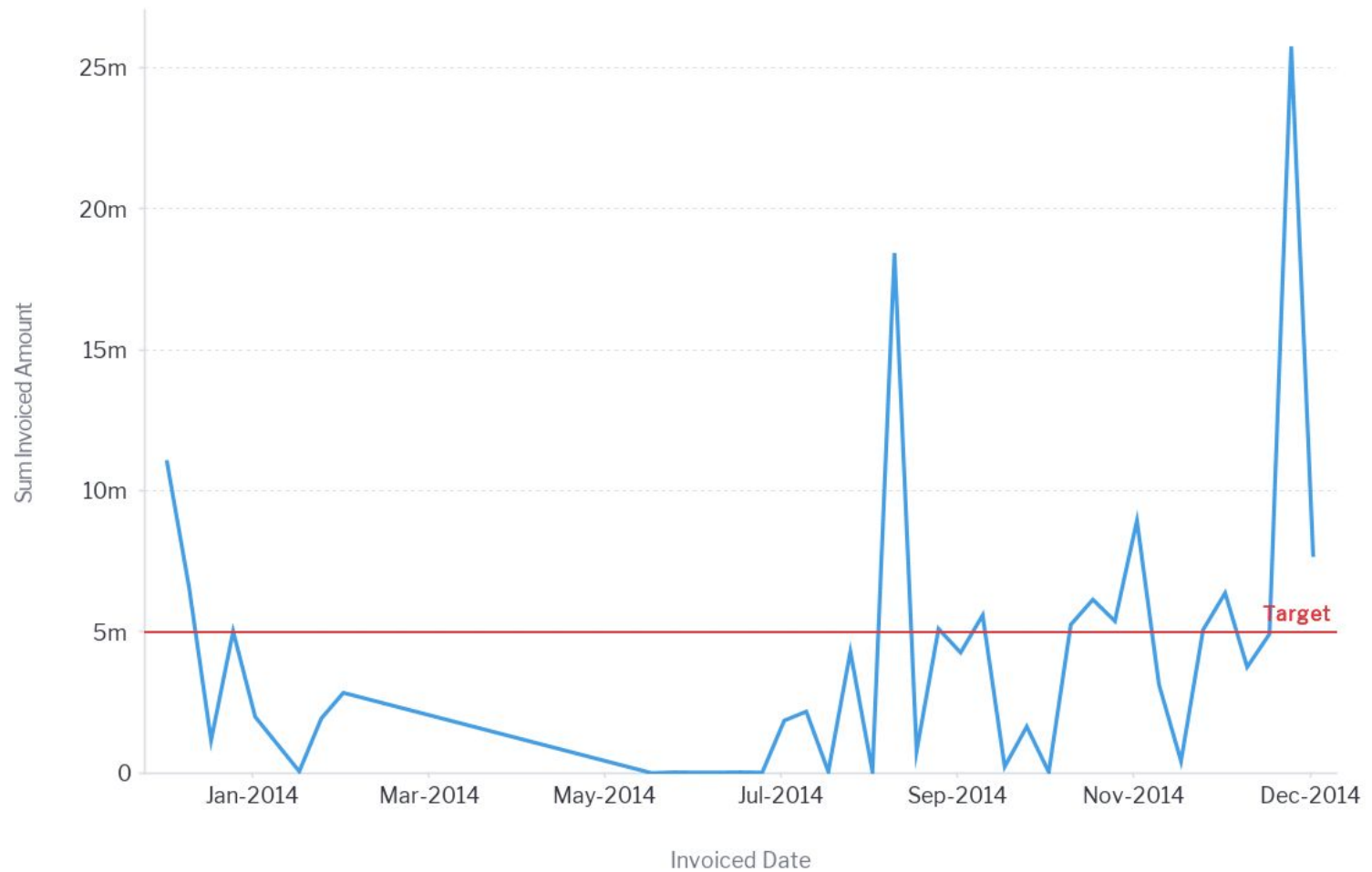
Other Examples

- Gantt Chart: Typically used in project management, Gantt charts are a bar chart depiction of timelines and tasks.
- Heat Map: A type of geospatial visualization in map form which displays specific data values as different colors (this doesn't need to be temperatures, but that is a common use).
- Highlight Table: A form of table that uses color to categorize similar data, allowing the viewer to read it more easily and intuitively.

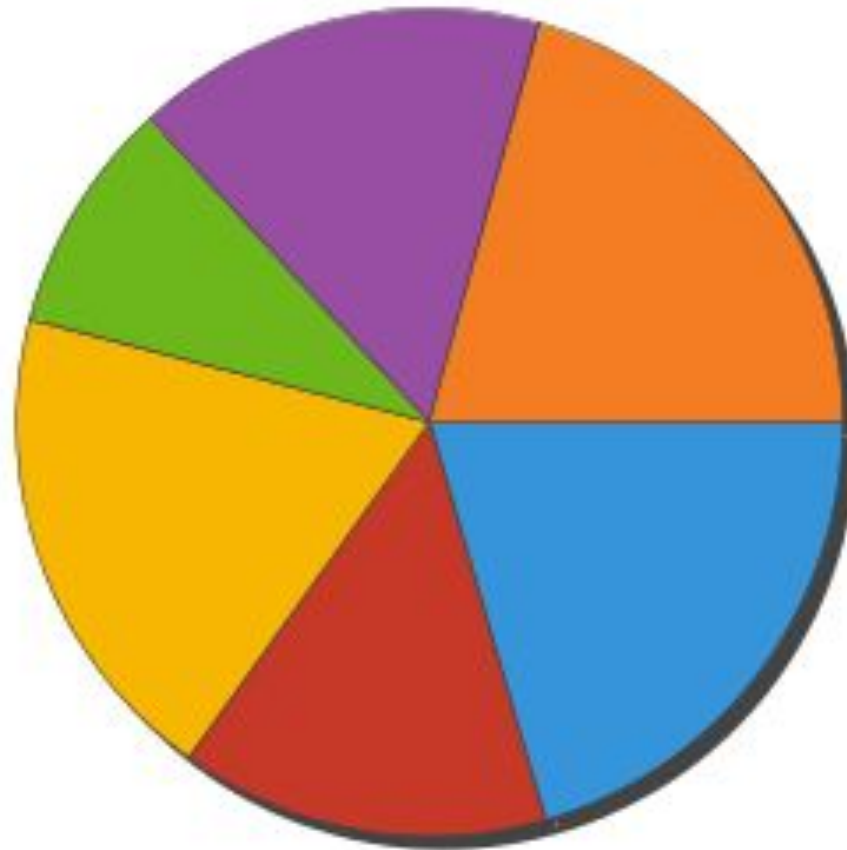
Other Examples

- Histogram: A type of bar chart that split a continuous measure into different bins to help analyze the distribution.
- Pie Chart: A circular chart with triangular segments that shows data as a percentage of a whole.
- Treemap: A type of chart that shows different, related values in the form of rectangles nested together.

Line Chart



Pie Chart

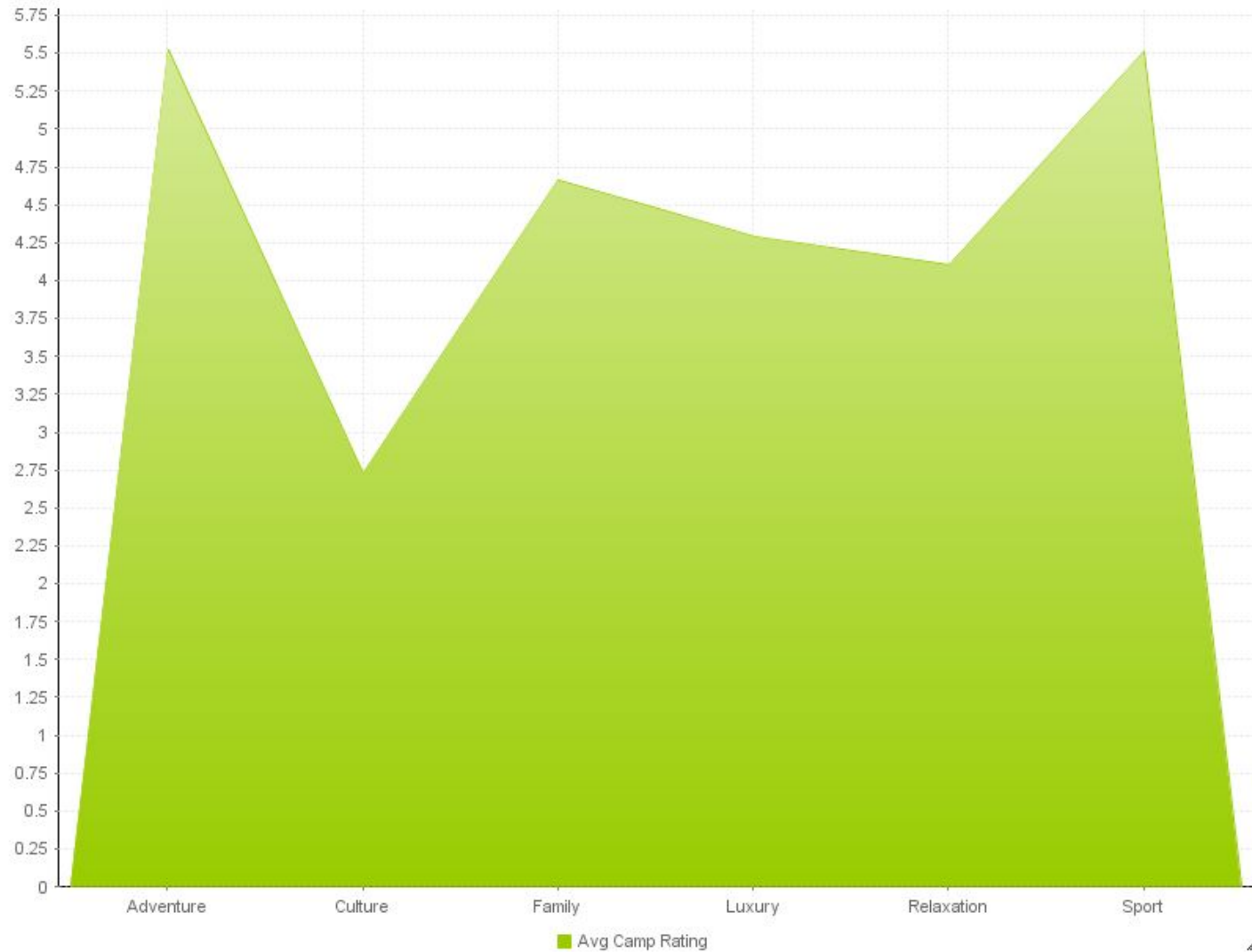


● Adventure ● Relaxation ● Family ● Culture ● Luxury ● Sport

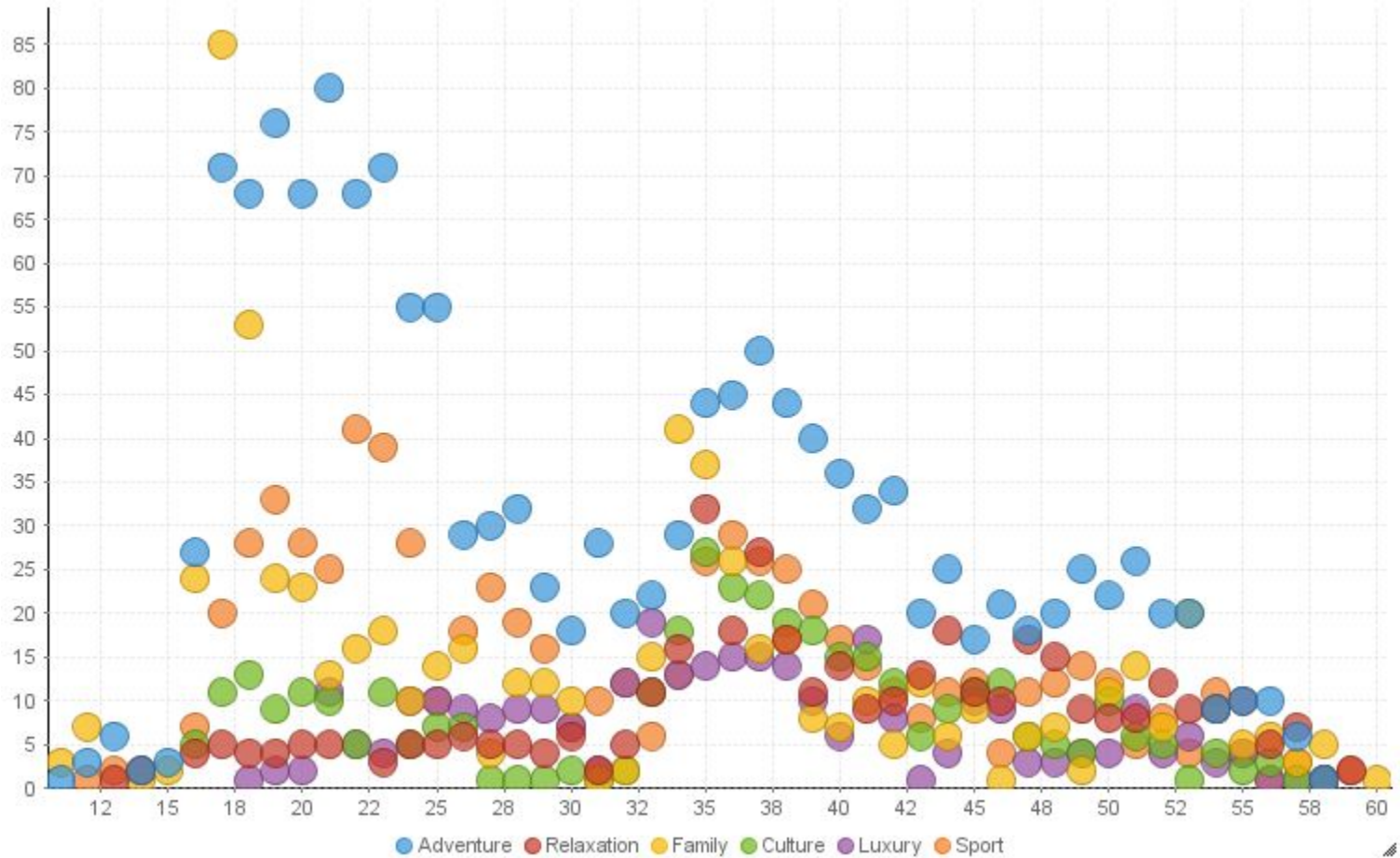
Bar Chart



Area Chart



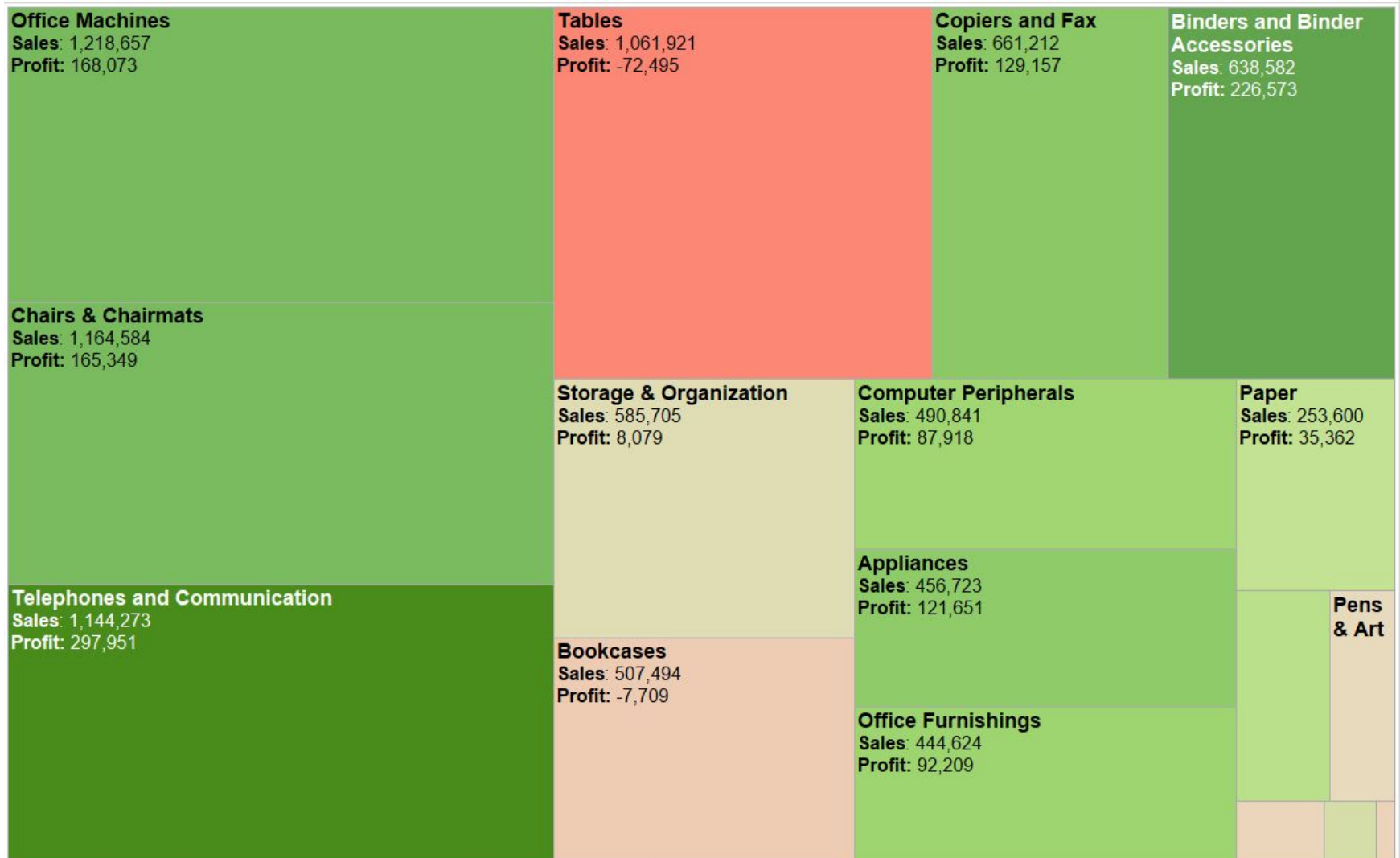
Scatter Chart



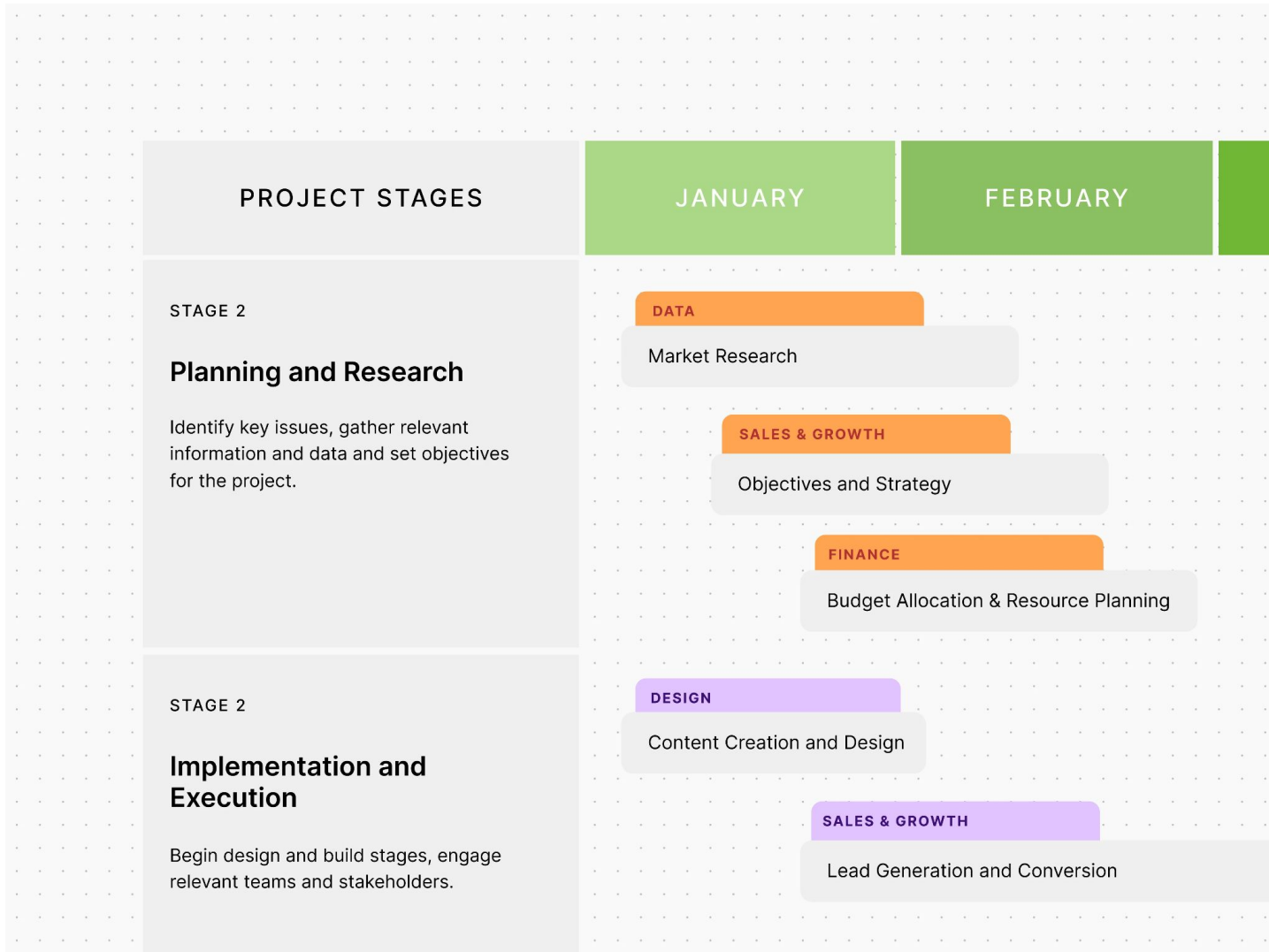
Bubble Chart



Treemap

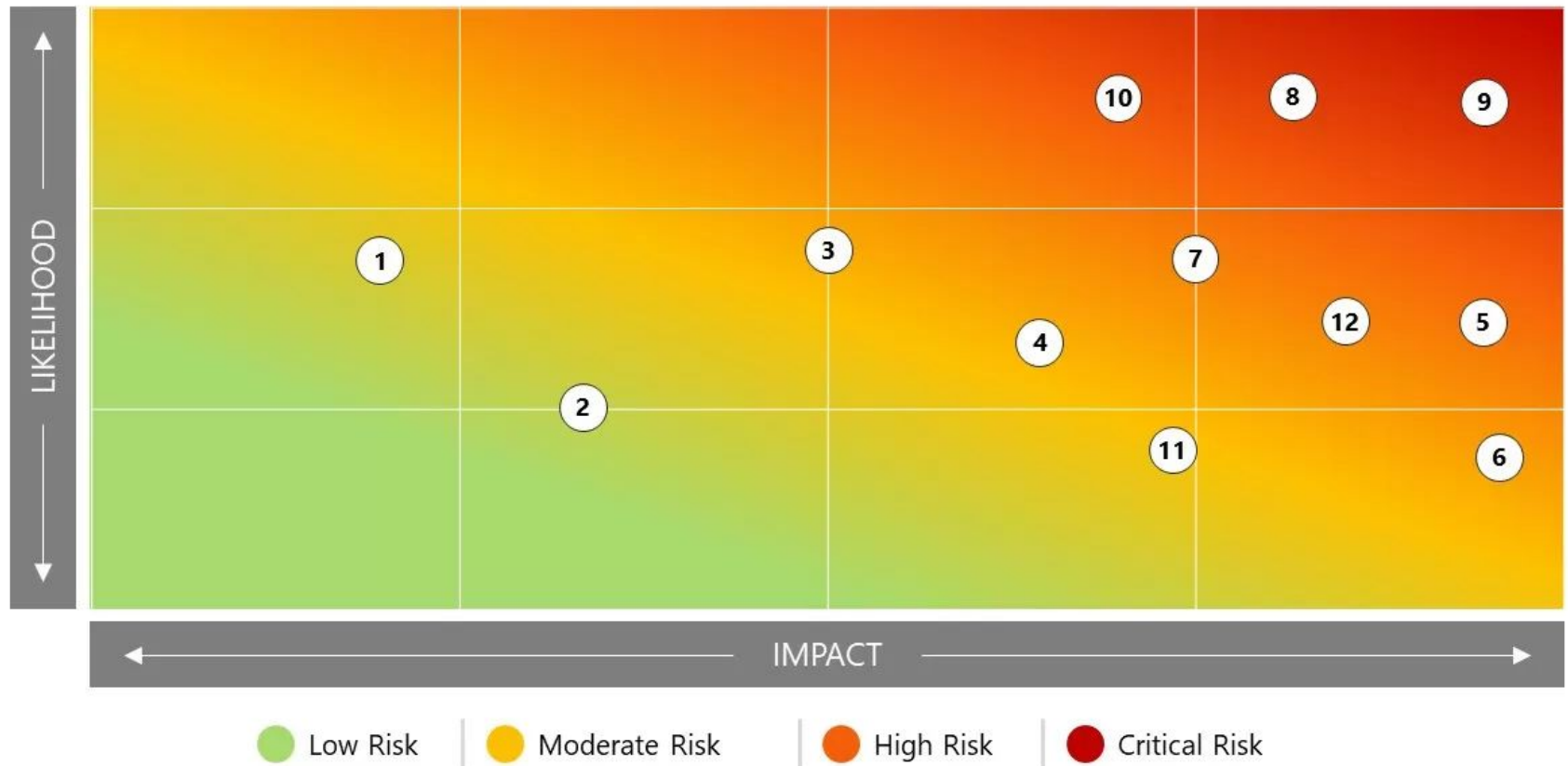


Gantt



Heat Map

Gradient Risk Matrix PowerPoint Template



Tools for Data Visualization



Thank You!